

ST PAUL 4 NE, AK, USA

WMO: 703090

Lat: 57.1575N

Lon: 170.2119W

Elev: 20

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 25711

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-0.4	4.4	-1.4	5.1	3.3	2.3	6.2	6.9	32.7	19.6	29.9	21.8	10.0	N/A	0.835

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	55.5	53.0	54.1	52.2	52.9	51.3	54.4	55.5	53.3	54.2	52.3	53.2	7.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.5	60.9	54.5	52.7	59.1	53.7	51.6	56.7	52.8	22.7	55.6	22.1	54.4	21.6	53.3	59.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.9	25.8	22.8	DB	-2.3	58.8	8.0	2.6	-8.1	60.7	-12.8	62.3	-17.3	63.7	-23.1	65.6	(4)
(5)			WB	0.1	56.0	8.3	1.7	-5.9	57.2	-10.7	58.2	-15.3	59.2	-21.3	60.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	24.8	25.7	23.3	28.9	35.4	42.1	46.9	49.0	45.9	39.6	33.7	29.5
	DBStd	10.83	8.65	9.62	10.16	6.76	4.83	3.58	3.31	2.79	3.93	4.72	5.40	6.67
	HDD50	5350	782	681	828	632	453	237	106	53	129	324	490	635
	HDD65	10784	1247	1101	1293	1082	918	686	560	495	573	789	940	1100
	CDD50	40	0	0	0	0	0	0	11	23	6	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.5	13.7	13.9	12.4	11.2	8.8	7.3	6.7	7.4	9.3	10.7	12.4	12.7
Precipitation	PrecAvg	23.3	1.7	1.3	1.2	1.2	1.1	1.3	1.9	2.9	2.8	3.0	2.9	2.2
	PrecMax	35.3	5.0	3.3	3.3	3.2	2.8	3.1	4.3	5.5	5.7	6.2	7.7	4.6
	PrecMin	9.8	0.3	0.0	0.1	0.1	0.1	0.0	0.4	0.2	0.6	0.9	0.9	0.6
	PrecStd	4.7	0.9	0.7	0.8	0.7	0.6	0.7	0.9	1.1	1.2	1.2	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	40.5	40.3	43.0	48.7	54.8	58.7	58.2	55.3	50.4	45.1	41.9
		MCWB	37.7	38.7	38.2	40.2	45.0	49.8	55.5	54.7	52.5	48.2	43.6	40.7
	2%	DB	38.7	38.9	38.6	40.7	45.5	51.1	55.4	55.8	53.5	49.3	44.0	40.5
		MCWB	37.0	37.1	36.8	38.7	42.8	47.5	53.2	53.4	51.2	47.5	42.8	39.3
	5%	DB	37.3	37.6	36.9	39.2	43.5	49.0	53.3	54.4	52.2	48.3	42.7	39.2
		MCWB	35.7	35.9	35.4	37.7	41.4	46.5	51.8	52.6	50.3	46.6	41.5	37.9
	10%	DB	35.9	36.6	35.0	37.5	42.0	47.5	51.7	53.3	51.1	46.8	41.3	37.9
		MCWB	34.5	35.0	33.6	36.3	40.2	45.6	50.5	52.0	49.4	45.2	39.8	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.0	39.9	39.3	41.2	45.4	51.2	56.2	55.6	54.1	49.5	44.5	41.7
		MCDB	39.4	40.4	39.9	42.4	47.2	54.3	58.1	57.2	54.7	49.8	44.7	41.9
	2%	WB	37.6	38.3	37.8	39.8	43.4	49.0	54.1	54.6	52.4	48.5	43.6	40.2
		MCDB	38.7	39.0	38.6	40.8	44.7	51.3	55.6	55.8	53.3	49.1	43.9	40.6
	5%	WB	36.7	37.0	36.4	38.7	42.2	47.7	52.7	53.7	51.3	47.6	42.5	38.7
		MCDB	37.8	37.9	37.5	39.7	43.4	49.4	53.9	54.5	52.2	48.5	43.1	39.4
	10%	WB	35.5	35.9	34.8	37.5	41.0	46.4	51.4	52.9	50.4	46.1	41.1	37.3
		MCDB	36.5	37.1	36.1	38.7	42.3	48.0	52.4	53.7	51.4	47.4	42.0	38.2
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.6	8.5	7.1	7.1	7.6	6.1	6.2	7.0	7.4	6.6	7.6
		MCDBR	5.9	5.7	7.5	7.3	9.8	10.4	8.4	7.8	7.3	5.8	6.6	7.4
		MCWBR	5.8	5.8	7.2	6.2	6.8	7.0	6.1	6.0	6.1	5.8	7.1	7.9
	5% WB	MCDBR	5.3	5.5	7.6	6.3	8.1	9.8	8.2	7.0	6.1	5.3	6.5	7.5
		MCWBR	5.4	6.0	7.3	5.5	6.3	7.2	6.7	5.7	5.3	5.7	7.3	7.7
Clear-Sky Solar Irradiance	taub	0.297	0.336	0.332	0.401	0.408	0.403	0.392	0.388	0.371	0.338	0.325	0.271	
	taud	2.211	2.204	2.286	2.096	2.201	2.306	2.401	2.411	2.358	2.318	2.144	2.320	
	Ebn at Noon	179	220	260	255	262	266	266	259	244	218	158	164	
	Edh at Noon	16	26	30	43	42	38	34	31	29	23	17	11	
All-Sky Solar Radiation	RadAvg	126	322	747	1122	1313	1358	1171	967	756	442	161	81	
	RadStd	15	36	122	128	96	97	88	98	59	28	18	11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page